RACE
Regular Analysis Through Continuous Exercise

Inverse Trigonometric Functions : RACE-01

Single Correct Question

1. If α, β are roots of the equation $6x^2 + 11x + 3 = 0$, then
(A) both $\cos^{-1}\alpha$ and $\cos^{-1}\beta$ exists
(B) both $\text{cosec}^{-1}\alpha$ and $\text{cosec}^{-1}\beta$ exists
(C) both $\cot^{-1}\alpha$ and $\cot^{-1}\beta$ exists
(D) none of these

MIT154

2. $\sin^{-1}\left(\sin\frac{33\pi}{7}\right)+\cos^{-1}\left(\cos\frac{46\pi}{7}\right)+\tan^{-1}\left(-\tan\frac{13\pi}{8}\right)+\cot^{-1}\left(\cot\left(-\frac{19\pi}{8}\right)\right)$ is equal to

(A) $\frac{11\pi}{7}$ (B) $\frac{9\pi}{7}$ (C) $\frac{13\pi}{7}$ (D) $\frac{17\pi}{7}$

MIT155

3. A function $g(x)$ satisfies the following conditions
(i) Domain of g is $(-\infty, \infty)$ (ii) Range of g is $[-1, 7]$
(iii) g has a period π and (iv) $g(2) = 3$
Then which of the following may be possible.
(A) $g(x) = 3 + 4\sin(n\pi + 2x - 4), n \in I$ (B) $g(x) = 3 - 8\sin(n\pi + 2x - 4), n \in I$
(C) $g(x) = 3 + 4\cos(n\pi + 2x - 4), n \in I$ (D) $g(x) = 3 - 8\cos(n\pi + 2x - 4), n \in I$

MIT156

4. Find the value of $\sin^{-1}(\cos(\sin^{-1}x)) + \cos^{-1}(\sin(\cos^{-1}x))$

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$ (C) π (D) 2π

MIT157

Matrix Match Type Question

5. Find the domain of each of the following functions :

	Column- I		**Column- II**
(P)	$f(x)=\sqrt{1-2x}+3\sin^{-1}\left(\frac{3x-1}{2}\right)$	(1)	ϕ
(Q)	$f(x)=2^{\sin^{-1}x}+\frac{1}{\sqrt{x-2}}$	(2)	$\left[-\frac{1}{3},\frac{1}{2}\right]$
(R)	$f(x)=\arccos\frac{2x}{1+x}$	(3)	$\{1,-1\}$
(S)	$f(x)=\sqrt{\cos(\sin x)}+\sin^{-1}\frac{1+x^2}{2x}$	(4)	$-1/3\le x\le 1$

MIT158

For More Join: @IITJEE_Advanced

6. Find the domain of definition of the following functions.
where {.} represents fractional part function

	Column- I		**Column- II**
(P)	$f(x)=\sin^{-1}\left(\frac{x-3}{2}\right)-\log_{10}(4-x)$	(1)	$(3/2, 2]$
(Q)	$f(x)=\frac{\sqrt{1-\sin x}}{\log_5(1-4x^2)}+\cos^{-1}(1-\{x\})$	(2)	$(-2, 2)-\{-1, 0, 1\}$
(R)	$f(x)=\sqrt{3-x}+\cos^{-1}\left(\frac{3-2x}{5}\right)+\log_6(2\lvert x\rvert-3)+\sin^{-1}(\log_2 x)$	(3)	$1\le x<4$
(S)	$f(x)=e^{\sin^{-1}\left(\frac{x}{2}\right)}+\tan^{-1}\left[\frac{x}{2}-1\right]+\ell n\left(\sqrt{x-[x]}\right)$	(4)	$x\in\left(-\frac{1}{2},\frac{1}{2}\right)-\{0\}$

MIT159

7. Find the range of each of the following functions:

	Column- I		**Column- II**
(P)	$f(x) = \ell n\ (\sin^{-1} x)$	(1)	$\left\{\frac{\pi}{4},\frac{3\pi}{4}\right\}$
(Q)	$f(x)=\sin^{-1}\left(\frac{\sqrt{3x^2+1}}{5x^2+1}\right)$	(2)	$[0, \pi]$
(R)	$f(x)=\cos^{-1}\left(\frac{(x-1)(x+5)}{x(x-2)(x-3)}\right)$	(3)	$(0, \pi/2]$
(S)	$f(x) = \sin^{-1} x + \tan^{-1} x + \sec^{-1} x$	(4)	$(-\infty, \ell n\ \pi/2]$

MIT160

Rankers Academy JEE

Integer Type Question

8. If $\sin^{-1} x + \sin^{-1} y = \pi$ and $x = ky$, then find the value of $(39)^{2k} + 5^k$.

MIT161

ANSWER KEY

Question	**1**	**2**	**3**	**4**	**5**	**6**
Answer	C	C	A	A	P→ 2; Q → 1 ; R → 4 ; S → 3	P → 3 ; Q → 4 ; R → 1 ; S → 2
Question	**7**				**8**	
Answer	P → 4 ; Q → 3 ; R → 2 ; S → 1				1526	

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Inverse Trigonometric Functions : RACE-02

Single Correct Question

1. If $\sin^{-1}x + \cot^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$, then x is equal to

(A) 0 (B) $\frac{1}{\sqrt{5}}$ (C) $\frac{2}{\sqrt{5}}$ (D) $\frac{\sqrt{3}}{2}$

MIT162

2. The value of $\cot\left(\text{cosec}^{-1}\frac{5}{3} + \tan^{-1}\frac{2}{3}\right)$ is equal to-

(A) $\frac{6}{17}$ (B) $\frac{3}{17}$ (C) $\frac{4}{17}$ (D) $\frac{5}{17}$

MIT163

3. If $\sin^{-1}\left(\frac{x}{5}\right) + \text{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$, then a value of x is-

(A) 1 (B) 3 (C) 4 (D) 5

MIT164

4. If $\pi \le x \le 2\pi$, then $\cos^{-1}(\cos x)$ is equal to

(A) x (B) $\pi - x$ (C) $2\pi + x$ (D) $2\pi - x$

MIT165

Rankers Academy JEE

5. If $\cos^{-1}x - \cos^{-1}\frac{y}{2} = \alpha$, then $4x^2 - 4xy\cos\alpha + y^2$ is equal to -

(A) $2\sin 2\alpha$ (B) 4 (C) $4\sin^2\alpha$ (D) $-4\sin^2\alpha$

MIT166

Multiple Correct Question

6. For what values of α, the equation, $(\sin^{-1}x)^3 + (\cos^{-1}x)^3 = \alpha\pi^3$ has no roots.

(A) $\left(-\infty, \frac{1}{32}\right)$ (B) $\left(\frac{7}{8}, \infty\right)$ (C) $\mathbb{R}$ (D) $\mathbb{R} - \{1\}$

MIT167

Matrix Match Type Question

7. Match the column:

	Column -I		**Column -II**
(P)	$\tan\left(\sin^{-1}\frac{3}{5} + \cot^{-1}\frac{3}{2}\right)$	(1)	$\frac{\sqrt{2}}{4}$
(Q)	$\cos\left(\sin^{-1}\frac{4}{5} + \cos^{-1}\frac{2}{3}\right)$	(2)	$2\sqrt{2}$
(R)	$\sin\left(\frac{1}{4}\text{arc}\sin\frac{\sqrt{63}}{8}\right)$	(3)	$\frac{17}{6}$
(S)	$\cot\left[\cos^{-1}\left(\frac{3}{4}\right) + \sin^{-1}\left(\frac{3}{4}\right) - \sec^{-1}3\right]$	(4)	$\frac{6-4\sqrt{5}}{15}$

MIT168

For More Join: @IITJEE_Advanced

8. Find the value of the following inverse trigonometric expressions :

	Column - I		**Column - II**
(P)	$\cos^{-1}(\cos 10)$	(1)	$2\pi - 6$
(Q)	$\tan^{-1}(\tan(-6))$	(2)	$4\pi - 10$
(R)	$\cot^{-1}(\cot(-8))$	(3)	$\frac{17\pi}{20}$
(S)	$\cos^{-1}\left(\frac{1}{\sqrt{2}}\left(\cos\frac{9\pi}{10}-\sin\frac{9\pi}{10}\right)\right)$	(4)	$3\pi - 8$

MIT169

Subjective Type Question

9. Find the set of values of 'a' for which the equation $2\cos^{-1}x = a + a^2(\cos^{-1}x)^{-1}$ posses a solution.

MIT170

Rankers Academy JEE

ANSWER KEY

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Answer	B	A	B	D	C	A,B	P → 3; Q → 4; R → 1; S → 2
Question	**8**				**9**		
Answer	P → 2; Q → 1; R → 4; S → 3				$a \in [-2\pi, \pi] - \{0\}$		

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Inverse Trigonometric Functions : RACE-03

Single Correct Question

1. Find all values of k for which there is a triangle whose angles have measure $\tan^{-1}\left(\frac{1}{2}\right), \tan^{-1}\left(\frac{1}{2}+k\right)$ and $\tan^{-1}\left(\frac{1}{2}+2k\right)$.

(A) 2 (B) 3 (C) $\frac{11}{4}$ (D) 5

MIT171

2. If x, y, z are in A.P. and $\tan^{-1} x$, $\tan^{-1} y$ and $\tan^{-1} z$ are also in A.P., then

(A) $x = y = z$ (B) $2x = 3y = 6z$ (C) $6x = 3y = 2z$ (D) $6x = 4y = 3z$

MIT172

3. Find the simplest value of $f(x) = arc\cos x + arc\cos\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right)$, $x \in \left(\frac{1}{2}, 1\right)$

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{9}$ (D) π

MIT173

Rankers Academy JEE

4. If $x \geq 0$ and $\theta = \sin^{-1} x + \cos^{-1} x - \tan^{-1} x$, then

(A) $\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{4}$ (B) $0 \leq \theta \leq \frac{\pi}{4}$ (C) $0 \leq \theta < \frac{\pi}{2}$ (D) $\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$

MIT174

5. If α is a real root of the equation $x^3 + 3x - \tan 2 = 0$, then $\cot^{-1}\alpha + \cot^{-1}\frac{1}{\alpha} - \frac{\pi}{2}$ can be equal to

(A) 0 (B) $\frac{\pi}{2}$ (C) π (D) $\frac{3\pi}{2}$

MIT175

6. Let $f : (-1, 1) \to B$, be a function defined by $f(x) = \tan^{-1}\frac{2x}{1-x^2}$, then f is both one-one and onto when B is the interval-

(A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (B) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (C) $\left(0, \frac{\pi}{2}\right)$ (D) $\left[0, \frac{\pi}{2}\right)$

MIT176

7. Let $\tan^{-1} y = \tan^{-1} x + \tan^{-1}\left(\frac{2x}{1-x^2}\right)$, where $|x| < \frac{1}{\sqrt{3}}$. Then a value of y is :

(A) $\frac{3x-x^3}{1+3x^2}$ (B) $\frac{3x+x^3}{1+3x^2}$ (C) $\frac{3x-x^3}{1-3x^2}$ (D) $\frac{3x+x^3}{1-3x^2}$

MIT177

For More Join: @IITJEE_Advanced

Multiple Correct Type Question

8. Find the value of $\sin\left(\tan^{-1}a+\tan^{-1}\frac{1}{a}\right);\ a\neq 0$

(A) 1 if $a>0$ (B) 0 (C) -1 if $a<0$ (D) 1

MIT178

9. Which of the following pair(s) are identical -

(A) $y=\tan(\cos^{-1}x);\ y=\frac{\sqrt{1-x^2}}{x}$ (B) $y=\tan(\cot^{-1}x);y=\frac{1}{x}$

(C) $y=\sin(\arctan x);y=\frac{x}{\sqrt{1+x^2}}$ (D) $y=\cos(\arctan x);y=\sin(\operatorname{arccot}x)$

MIT179

Matrix Match Type Question

10. Match the column :

	Column-I		**Column-II**
(P)	$\tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\left(\frac{3}{5}\right)-\tan^{-1}\left(\frac{8}{19}\right)$	(1)	$\tan^{-1}\left(\frac{1}{2}\right)$
(Q)	$\tan^{-1}\left(\frac{2}{11}\right)+\tan^{-1}\left(\frac{7}{24}\right)$	(2)	$\frac{\pi}{4}$
(R)	$2\cos^{-1}\frac{3}{\sqrt{13}}+\cot^{-1}\frac{16}{63}+\frac{1}{2}\cos^{-1}\frac{7}{25}$	(3)	$\frac{\pi}{6}$
(S)	$\text{arc}\cos\sqrt{\frac{2}{3}}-arc\cos\frac{\sqrt{6}+1}{2\sqrt{3}}$	(4)	π

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MIT180

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	C	A	B	D	C	B	C	A,C	A,B,C,D
Question	10								
Answer	P → 2; Q → 1; R → 4; S → 3								

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Inverse Trigonometric Functions : RACE-04

Single Correct Question

1. The numerical value of $\cot\left(2\sin^{-1}\frac{3}{5}+\cos^{-1}\frac{3}{5}\right)$ is

(A) $\frac{-4}{3}$ (B) $\frac{-3}{4}$ (C) $\frac{3}{4}$ (D) $\frac{4}{3}$

MIT181

2. Simplify $\tan\left\{\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right)+\frac{1}{2}\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right\}$, if $x > y > 1$.

(A) $\frac{1-xy}{x-y}$ (B) $\frac{1+xy}{x-y}$ (C) $\frac{1+xy}{x+y}$ (D) $\frac{1-xy}{x+y}$

MIT182

3. Solve $\tan^{-1}\frac{x-1}{x+1}+\tan^{-1}\frac{2x-1}{2x+1}=\tan^{-1}\frac{23}{36}$

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{7}{3}$ (D) $\frac{3}{7}$

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MIT183

4. Solve $2\tan^{-1}x=\cos^{-1}\frac{1-a^2}{1+a^2}-\cos^{-1}\frac{1-b^2}{1+b^2}$ $(a>0, b>0)$

(A) $x=\frac{a-b}{1+ab}$ (B) $x=\frac{a+b}{1-ab}$ (C) $x=\frac{a+b}{1+ab}$ (D) $x=\frac{a-b}{1-ab}$

MIT184

Multiple Correct Type Question

5. $\tan^{-1}x, x<0$ is equal to

(A) $-\pi+\cot^{-1}\frac{1}{x}$ (B) $\sin^{-1}\frac{x}{\sqrt{1+x^2}}$ (C) $-\cos^{-1}\frac{1}{\sqrt{1+x^2}}$ (D) $\cos^{-1}\frac{1}{\sqrt{1+x^2}}$

MIT185

6. $\cos^{-1}x, -1<x<0$ is equal to

(A) $\sec^{-1}\frac{1}{x}$ (B) $\pi-\sin^{-1}\sqrt{1-x^2}$ (C) $\pi+\tan^{-1}\frac{\sqrt{1-x^2}}{x}$ (D) $\cot^{-1}\frac{x}{\sqrt{1-x^2}}$

MIT186

For More Join: @IITJEE_Advanced

Matrix Match Type Question

7. Prove that

	Column – I		**Column – II**
(P)	$\sin^{-1}\left(\frac{3}{5}\right)+\sin^{-1}\left(\frac{8}{17}\right)$	(1)	$\frac{\pi}{2}$
(Q)	$\tan^{-1}\frac{3}{4}+\sin^{-1}\frac{5}{13}$	(2)	$\frac{\pi}{4}$
(R)	$2[\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)+\cot^{-1}3]$	(3)	$\sin^{-1}\frac{77}{85}$
(S)	$\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{5}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{8}\right)$	(4)	$\cos^{-1}\frac{33}{65}$

MIT187

Integer Type Question

8. If $\tan^{-1}x+\cot^{-1}\frac{1}{y}+2\tan^{-1}z=\pi$, then $x+y+\mu z=xz^2+yz^2+\lambda xyz$. Find $\lambda+\mu$

MIT188

9. If $a>0, b>0, \tan^{-1}\left(\frac{a}{x}\right)+\tan^{-1}\left(\frac{b}{x}\right)=\frac{\pi}{2}$ then $x=(ab)^{\frac{1}{n}}$ find $n(n\in N)$

MIT189

Rankers Academy JEE

Subjective Type Question

10. Prove the following results :

(i) $2\tan^{-1}\left(\frac{1}{5}\right)+\tan^{-1}\left(\frac{1}{8}\right)=\tan^{-1}\left(\frac{4}{7}\right)$

(ii) $2\sin^{-1}\left(\frac{3}{5}\right)-\tan^{-1}\left(\frac{17}{31}\right)=\frac{\pi}{4}$

MIT190

ANSWER KEY

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Answer	B	B	B	A	A,B,C	A,B,C,D	P → 3; Q → 4; R → 1 ; S → 2
Question	**8**	**9**					
Answer	4	2					

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Inverse Trigonometric Functions : RACE-05

Single Correct Question

1. Solve the inequality $\tan^{-1} x > \cot^{-1} x$.

(A) $(1, \infty)$ (B) $(-2, 0)$

(C) $(-\infty, 1)$ (D) None of these

MIT191

2. Complete solution set of inequation $(\cos^{-1} x)^2 - (\sin^{-1} x)^2 > 0$, is

(A) $\left[0, \frac{1}{\sqrt{2}}\right)$ (B) $\left[-1, \frac{1}{\sqrt{2}}\right)$

(C) $(-1, \sqrt{2})$ (D) none of these

MIT192

3. All x satisfying the inequality $(\cot^{-1} x)^2 - 7(\cot^{-1} x) + 10 > 0$, lie in the interval :-

(A) $(-\infty, \cot 5) \cup (\cot 4, \cot 2)$ (B) $(\cot 5, \cot 4)$

(C) $(\cot 2, \infty)$ (D) $(-\infty, \cot 5) \cup (\cot 2, \infty)$

MIT193

Rankers Academy JEE

4. Solve for x : $\sin^{-1}\left(\sin\left(\frac{2x^2+4}{1+x^2}\right)\right) < \pi - 3$.

(A) $x \in (-2, -1)$ (B) $x \in (-1, 1)$ (C) $x \in (2, 4)$ (D) $x \in (-3, -1)$

MIT194

5. The solution of the equation $\sin^{-1}\left(\tan\frac{\pi}{4}\right) - \sin^{-1}\left(\sqrt{\frac{3}{x}}\right) - \frac{\pi}{6} = 0$ is

(A) $x = 2$ (B) $x = -4$ (C) $x = 4$ (D) $x = 3$

MIT195

6. Number of solutions of the equation $\cot^{-1}\sqrt{4-x^2} + \cos^{-1}(x^2-5) = \frac{3\pi}{2}$ is :

(A) 2 (B) 4 (C) 6 (D) 8

MIT196

Multiple Correct Type Question

7. Find all the positive integral solutions (x, y) of, $\tan^{-1} x + \cos^{-1}\frac{y}{\sqrt{1+y^2}} = \sin^{-1}\frac{3}{\sqrt{10}}$.

(A) (1,2) (B) (2, 3) (C) (2, 7) (D) (1, 4)

MIT197

For More Join: @IITJEE_Advanced

Matrix Match Type Question

8. Solve for x :

	Column – I		**Column – II**
(P)	$\sin^{-1}x+\sin^{-1}2x=\frac{\pi}{3}$	(1)	$x=3$
(Q)	$\tan^{-1}\frac{1}{1+2x}+\tan^{-1}\frac{1}{1+4x}=\tan^{-1}\frac{2}{x^2}$	(2)	$x=\frac{1}{2}\sqrt{\frac{3}{7}}$
(R)	$\tan^{-1}(x-1)+\tan^{-1}(x)+\tan^{-1}(x+1)=\tan^{-1}(3x)$	(3)	$\left[\frac{\sqrt{3}}{2},1\right]$
(S)	$3\cos^{-1}x=\sin^{-1}\left(\sqrt{1-x^2}(4x^2-1)\right)$	(4)	$x=0,\frac{1}{2},-\frac{1}{2}$

MIT198

Subjective Type Question

9. Solve the following inequalities:

(i) $\text{arc cot}^2 x - 5 \text{ arc cot } x + 6 > 0$

(ii) $\text{arc sin } x > \text{arc cos } x$

(iii) $\tan^2 (\text{arc sin } x) > 1$

MIT199

10. Solve the following system of inequalities :

$4 \text{ arc tan}^2 x - 8 \text{ arc tan } x + 3 < 0$ & $4 \text{ arc cot } x - \text{arc cot}^2 x - 3 \geq 0$

MIT200

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	A	B	C	B	C	A	A,C	P → 2; Q → 1; R → 4; S → 3

9. (i) $(\cot 2, \infty) \cup (-\infty, \cot 3)$ (ii) $\left(\frac{\sqrt{2}}{2}, 1\right]$ (iii) $\left(\frac{\sqrt{2}}{2}, 1\right) \cup \left(-1, -\frac{\sqrt{2}}{2}\right)$

10. $\left(\tan\frac{1}{2}, \cot 1\right]$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Inverse Trigonometric Functions : RACE-06

Single Correct Question

1. $\tan^{-1}\frac{1}{x^2+x+1}+\tan^{-1}\frac{1}{x^2+3x+3}+\tan^{-1}\frac{1}{x^2+5x+7}+\tan^{-1}\frac{1}{x^2+7x+13}$ .... upto n terms.

(A) $\tan^{-1}(x+n)-\tan^{-1}x$ (B) $\tan^{-1}(x+n-1)-\tan^{-1}x$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

MIT201

2. $\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{2}{9}+.....+\tan^{-1}\frac{2^{n-1}}{1+2^{2n-1}}+$ .....upto infinite terms

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$ (C) $\frac{\pi}{4}$ (D) π

MIT202

3. $\sin^{-1}\frac{1}{\sqrt{2}}+\sin^{-1}\frac{\sqrt{2}-1}{\sqrt{6}}+.....+\sin^{-1}\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}}+$ ..... upto infinite terms

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$ (C) $\frac{\pi}{4}$ (D) π

Rankers Academy JEE

MIT203

4. $\cot^{-1}7+\cot^{-1}13+\cot^{-1}21+\cot^{-1}31+.....$ to n terms.

(A) $\text{arccot}\left[\frac{2n+5}{2n}\right]$ (B) $\text{arccot}\left[\frac{2n+6}{n}\right]$ (C) $\text{arccot}\left[\frac{2n+5}{n}\right]$ (D) $\frac{\pi}{4}$

MIT204

5. $\sin^{-1}\frac{1}{\sqrt{5}}+\sin^{-1}\frac{1}{\sqrt{65}}+\sin^{-1}\frac{1}{\sqrt{325}}+.....+\sin^{-1}\frac{1}{\sqrt{4n^4+1}}+.....\infty$ terms.

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$ (C) $\frac{\pi}{4}$ (D) π

MIT205

6. If $x\in(0,1)$ and $f(x)=\sec\left\{\tan^{-1}\left(\frac{\sin(\cos^{-1}x)+\cos(\sin^{-1}x)}{\cos(\cos^{-1}x)+\sin(\sin^{-1}x)}\right)\right\}$, then $\sum_{r=2}^{10}f\left(\frac{1}{r}\right)$ is

(A) 54 (B) 52 (C) 108 (D) 27

MIT206

7. Find the value of $3\sum_{n=1}^{\infty}\left\{\frac{1}{\pi}\sum_{k=1}^{\infty}\cot^{-1}\left(1+2\sqrt{\sum_{r=1}^{k}r^3}\right)\right\}^n$

(A) 2 (B) 1 (C) 3 (D) $\frac{1}{2}$

MIT207

For More Join: @IITJEE_Advanced

Matrix Match Type Question

8. Solve for x :

	Column – I		**Column – II**
(P)	$\cos(2\sin^{-1}x) = \frac{1}{3}$	(1)	$\pm\frac{1}{\sqrt{2}}$
(Q)	$\cot^{-1}x + \tan^{-1}3 = \frac{\pi}{2}$	(2)	$x = \frac{1}{2}$
(R)	$\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$	(3)	$\pm\frac{1}{\sqrt{3}}$
(S)	$\sin^{-1}x + \sin^{-1}2x = \frac{2\pi}{3}$	(4)	$x = 3$

MIT208

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	A	C	A	C	C	A	B	P → 3; Q → 4; R → 1; S → 2

For More Join: @IITJEE_Advanced